Listing 3.1 A simple example of how a continuously varying voltage is quantized to a discrete value by the process of analog to digital conversion.

```
def adc_read(voltage, bits=10, vref=3.3): #1
  max_val = 2**bits - 1
  return int((voltage / vref) * max_val) #2

def to_moisture(adc_value, dry=850, wet=400):
  if adc_value >= dry: return 0
  if adc_value <= wet: return 100
  return 100 * (dry - adc_value) / (dry - wet)

print(f"{to_moisture(adc_read(1.5)):5.1f}%")
```

#1 ADC function takes in in the voltage, the number of bits, and the maximum possible voltage.
#2 Conversion formula from analog to digital.

3.1.1 Key sensor characteristics

Selecting the right sensors for your digital twin is rarely as simple as finding those that measure the parameters you're interested in. You must understand the key characteristics that determine how well a sensor will perform in your specific environment and use case, and evaluate your sensor choice against these.

- *Temporal resolution* refers to how frequently the sensor can provide measurements. A manufacturing line might need temperature readings every second, while a building's occupancy sensor might only need updates every few minutes. Match the sampling rate to your system's dynamics and decision-making needs.
- *Measurement range and sensitivity* defines the limits every sensor operates within. A humidity sensor designed for indoor use might fail in an industrial dryer where localized humidity spikes beyond normal ranges. Similarly, sensitivity determines the smallest change the sensor can detect, which is important when monitoring small changes and gradual trends.
- *Accuracy* is how close your measurement is to the true value, while *precision* is how repeatable your measurements are. A sensor might consistently read 22.3°C when the actual temperature is 25.0°C, so it is precise but not accurate. For digital twins, you often need both, but